SOKOINE UNIVERSITY OF AGRICULTURE
School of Engineering and Technology

Course: ABE 721 - Agricultural Systems Modelling and Simulation
Instructor: Kadeghe Fue, PhD, P.Eng(T)

ASSIGNMENT 4
Module 4: Model Validation and Analysis

Instructions:

- This assignment bridges the gap between creating a functional model and proving its statistical reliability and stability under chaotic, real-world agricultural conditions.
- Carefully read the ”Significance” statement for each question to understand the practical engineering application of the theory.
- Show all logical steps in your mathematical derivations and interpretations.
- **Programming Requirement:** For questions that require computational implementation, you must submit your complete, well-commented source code (Python, MATLAB, or R) along with the exact results (console transcripts and any generated plots).

Tier 1: Simple (Conceptual Foundations)

Question 1.1: Verification vs. Validation and Metrics

Significance of Question: *Differentiating between model verification and validation ensures we are not just solving equations correctly, but that we are solving the right equations that accurately mirror physical reality.*

Prompt: Briefly distinguish between the concepts of Verification and Validation in systems modelling. Furthermore, if you calculate a Nash-Sutcliffe Efficiency (NSE) of -0.45 for your hydrological model, what does this specific value mathematically and practically imply about your model’s predictive power?

Question 1.2: System Dynamics Feedback Loops

Significance of Question: *Properly identifying the polarity of causal links and loops is the absolute first step in translating qualitative system behavior into rigorous, predictable differential equations.*

Prompt: In the context of Causal Loop Diagrams (CLDs), define what constitutes a **Reinforcing (R)** loop versus a **Balancing (B)** loop based on link polarities. Provide one short agricultural example of a Reinforcing loop (e.g., relating to pest populations).

Tier 2: Intermediate (Application and Logic)

Question 2.1: Stochastic Probability Distributions

Significance of Question: *Deterministic models completely ignore the chaotic reality of farming. Utilizing the correct probability density function captures extreme biological and weather failure risks.*

Prompt:

a) Why is the **Log-Normal distribution** generally preferred over the standard Normal (Gaussian) distribution for modelling daily rainfall or streamflow in stochastic agricultural models?

b) Explain the conceptual and computational advantage of **Latin Hypercube Sampling (LHS)** over standard Monte Carlo sampling when trying to analyze extreme, rare events (like a 1-in-100 year flood).

Question 2.2: Jacobian Stability and Phase Planes

Significance of Question: *Analyzing the eigenvalues of a Jacobian matrix allows engineers to mathematically guarantee whether a biological ecosystem will reach a stable, resilient equilibrium or spiral into chaotic collapse.*

Prompt: Assume you have mathematically linearized a 2D agricultural system around its equilibrium point $\mathbf{x}^*$. The resulting Jacobian matrix **J** yields the following two complex eigenvalues:

$$\lambda_1 = -0.15 + 1.2i, \quad \lambda_2 = -0.15 - 1.2i$$

Based on the Stability Theorem, is this equilibrium point locally asymptotically stable or unstable? Furthermore, due to the presence of the imaginary component ($\pm 1.2i$), how will the state variables physically behave as they approach (or leave) this equilibrium over time?

Tier 3: Hard (Synthesis and Formulation)

Question 3.1: SPAC and Penman-Monteith Analysis

Significance of Question: *Modelling the Soil-Plant-Atmosphere Continuum allows for the precision scheduling of automated irrigation systems based on strict thermodynamic physics rather than human guesswork.*

Prompt: The flow of water (q) through the Soil-Plant-Atmosphere Continuum (SPAC) relies heavily on an electrical resistance analogy (Ohm's Law).

a) Write the mathematical flow equation showing how water moves from the soil potential (Ψ_{soil}) all the way to the atmospheric potential (Ψ_{atm}) through the various resistances.

b) In the context of the FAO-56 Penman-Monteith Evapotranspiration equation, explain the physical and biological significance of the **Bulk Surface Stomatal Resistance (r_s)** term. What physically happens to this parameter, and the overall ET equation, when the plant enters severe drought stress?

Question 3.2: Programming Validation Metrics and Monte Carlo

Significance of Question: *Hard-coding statistical validation metrics and basic stochastic simulations is a mandatory foundational skill for modern data-driven agricultural software engineering.*

Prompt (Programming Task): You must write a script in Python, MATLAB, or R to accomplish the following two separate tasks:

Part A (Validation): You have recorded the following arrays of crop yield data (tons/ha):

- Observed Yields: $[4.2, 4.5, 3.8, 5.1, 4.0, 4.8]$
- Simulated Yields: $[4.0, 4.6, 3.5, 4.9, 4.1, 4.5]$

Write functions/code to calculate and print the **RMSE** (Root Mean Square Error), **MAE** (Mean Absolute Error), and **NSE** (Nash-Sutcliffe Efficiency) for this dataset.

Part B (Stochastic Monte Carlo): Write a script to generate a simple Monte Carlo simulation consisting of $N = 10,000$ iterations. In each iteration, randomly sample a daily rainfall value from a **Log-Normal distribution** with a mean (μ) of 1.5 and a standard deviation (σ) of 0.6. Generate and display a histogram plot of the resulting 10,000 simulated rainfall values.

Submission Requirements: Submit your fully commented source code, the exact console text output showing your three validation metrics, and the generated histogram plot.

Please submit your work to my email kadeghe.fue@sua.ac.tz by one week after completion of Module 4. Do not send this work to kadefue@sua.ac.tz.